Popular Protocols in TCP/IP Networking

Two most popular and widely adopted networking models are:

- ISO OSI model
- TCP/IP model

TCP/IP model

This networking model defines four functional layers of networking for computing devices:

- Application
- Transport
- Internet
- Network Access

Functionalities of the four layers of networking:

Application layer: It is about encoding and decoding the data in end-user context and business context. Various protocols make it possible; some of them are: Telnet, SMTP, POP3, FTP, NTP, HTTP, SNMP, DNS, SSH

Transport layer: It takes care of the transfer of data across devices in the form of stream with markers (bit sequences) indicating the start of individual data packets, their end and optionally error detection-and-correction codes. The protocols for this layer are: TCP, UDP

Internet layer: This layer dictates how the devices in a network establish connections together. It is mainly concerned with their addressing schemes. Protocols are: IP, ICMP, ARP, DHCP

Network Access layer: This layer maintains various options and strategies to set the network up among devices. To choose among and configure network topologies, these protocol are helpful: Ethernet, PPP, ADSL

Acronyms and Abbreviations

ISO: International Standards Organization

OSI: open source initiative

IP: internetwork protocol / internetworking pProtocol

Telnet: telecommunication networking

SMTP: simple mail transfer protocol

POP3: post office protocol 3

FTP: file transfer protocol

NTP: network time protocol

HTTP: hypertext transfer protocol

SNMP: simple network management protocol

DNS: domain naming service

SSH: secure shell

TCP: transmission control protocol

UDP: user datagram protocol

ICMP: internet control message protocol

ARP: address resolution protocol

DHCP: dynamic host control protocol

Ethernet: ether network

PPP: point-to-point protocol (P2P protocol)

ADSL: asymmetrical digital subscriber line

Note

Small initial letters are use in the full form for the enhanced to eyes.

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